

Policy Bill Budget AI: Union Budget Intelligence Assistant for India

1. Project Title

Policy Bill Budget AI – An AI-Powered Union Budget Analysis and Policy Intelligence System

2. Project Overview

Policy Bill Budget AI is an AI-powered system designed to analyze India's Union Budget, Finance Bills, government policies, and sector-wise budget allocations. The system will help users understand how government revenue is collected, how taxpayer money is spent, which sectors receive priority, and how budget allocations change over time.

The project will not be limited to simple graphs or dashboards. Instead, it will work as a **budget intelligence assistant** that can summarize budget documents, compare current and previous budgets, detect trends, identify unusual allocation changes, and explain the impact of policies in simple language.

The system will also compare selected Indian budget sectors with foreign budgets, especially the US budget, to understand differences in spending priorities across areas such as finance, education, healthcare, private sector support, agriculture, and defense.

3. Problem Statement

Government budget documents are lengthy, complex, and difficult for students, taxpayers, farmers, startups, and general citizens to understand. Although official government websites provide budget documents, graphs, and year-wise comparisons, they often do not explain the real-world impact of budget changes in a simple and interactive way.

Users may still struggle to answer questions such as:

- How is taxpayer money being spent?
- Which sectors received more or less funding compared to last year?
- What new reforms were introduced in the budget?
- How does India's spending on education, health, defense, or agriculture compare with previous years?
- Are government priorities in budget speeches matching actual fund allocation?
- How does India's budget allocation compare with countries like the US?

This project aims to solve these problems using machine learning, data visualization, natural language processing, and comparative policy analysis.

4. Project Objective

The main objective of this project is to build a minimal working AI-based Union Budget analysis system and then gradually add advanced features.

The system will:

- Summarize Union Budget PDFs and Finance Bills.
- Compare current budget data with previous years.
- Analyze sector-wise impact in education, healthcare, defense, agriculture, infrastructure, finance, and private sector support.
- Visualize government earnings and expenditure using charts.
- Detect patterns, trends, and unusual budget changes.
- Explain “who benefits” and “who may be affected” by a budget decision.
- Compare India’s sector-wise expenditure with selected foreign budgets such as the US Federal Budget.
- Predict future budget trends based on historical allocation data.
- Compare policy priorities mentioned in budget speeches with actual fund allocation.

5. What the Project Will Solve

This project will solve the following problems:

1. Understanding how taxpayer money is spent

The system will explain how income tax, corporate tax, GST, borrowing, and other revenue sources are used in different sectors.

2. Budget visualization

It will create visual charts such as pie charts, bar graphs, line graphs, and allocation-vs-spending graphs.

3. Sector-wise budget comparison

The system will compare money allocated to sectors such as education, health, defense, agriculture, infrastructure, finance, and private sector support.

4. GDP and previous year comparison

It will compare budget allocation with GDP percentage and previous year allocation.

5. Pattern and trend detection

The system will identify trends such as income tax relief, increase in government expenditure, new reforms, subsidy changes, and sector-wise budget growth.

6. Government earning analysis

It will compare estimated government earnings from sources such as GST, income tax, corporate tax, borrowing, customs duty, and non-tax revenue.

7. India vs US budget comparison

The system will compare expenditure patterns between India and the US in selected sectors such as education, healthcare, finance, private sector support, and defense.

8. Expected vs actual spending analysis

It will compare actual spending with recommended or benchmark values. For example, education spending can be compared with a 6% benchmark, while health expenditure can be compared with a 5% benchmark.

6. Core Features

Feature 1: Bill PDF Summarizer

Users will be able to upload Union Budget PDFs, Finance Bills, budget speeches, or policy documents. The system will summarize lengthy documents into simple and understandable points.

Output examples:

- Key highlights of the bill
- Important tax changes
- New schemes introduced
- Subsidy changes
- Sector-wise budget priorities

Feature 2: Natural Language Question Answering

Users can ask questions in simple English, such as:

- "How much money was allocated to healthcare this year?"
- "Which sector received the highest increase?"
- "What are the major tax reforms in this budget?"
- "How much has defense spending increased from last year?"

The system will answer using extracted budget data and document context.

Feature 3: Previous Budget Comparison

The system will compare the current Union Budget with previous years.

It will compare:

- Current year vs previous year
- Current year vs 5-year average
- Sector-wise allocation changes
- Capital expenditure vs revenue expenditure
- Fiscal deficit trends
- Tax revenue vs non-tax revenue

Feature 4: Sector-Wise Impact Analysis

The system will provide separate analysis for major sectors:

- Education
- Healthcare
- Defense
- Agriculture
- Infrastructure
- Finance
- Private sector support
- MSME
- Railways
- Pollution control and environment

For each sector, the system will show:

- Budget allocated
- Percentage increase or decrease from last year
- Share of total budget
- New policies or schemes
- Possible beneficiaries
- Possible affected groups

Feature 5: Budget Allocation Charts

The project will include a separate visualization section.

Important visualizations will include:

- Pie chart of government earnings from income tax, GST, corporate tax, borrowing, and other sources
- Bar graph comparing budget estimates for financial years 2025–26 and 2026–27
- Line graph for corporate tax and personal income tax trends
- Pie chart for budget allocation across defense, central sector schemes, state share of taxes, and interest payments
- Bar graph for health expenditure across financial years
- Bar graph for capital expenditure across financial years
- Bar graph for pollution control expenditure across financial years
- Allocation vs actual spending graph

Feature 6: Who Benefits and Who May Be Affected

The system will explain budget impact in public-friendly language.

Example:

- Students may benefit if education allocation increases.
- Farmers may benefit if agriculture schemes receive higher funding.

- Startups may benefit if tax relief or credit schemes are introduced.
- Taxpayers may benefit if income tax slabs are revised.
- Certain sectors may be affected if subsidies or allocations are reduced.

Feature 7: Scheme-Level Allocation Comparison

The system will compare important schemes across different years.

Examples:

- PM-KISAN allocation trend
- Ayushman Bharat allocation trend
- Education schemes allocation trend
- Defense modernization allocation trend
- Infrastructure schemes allocation trend

This will help identify which schemes are receiving more or less priority.

Feature 8: Anomaly Detection

The system will detect unusual increases or decreases in budget allocation.

Example output:

- "Defense allocation increased significantly compared to the 5-year average."
- "Health expenditure growth is lower than the previous year."
- "Pollution control allocation shows an unusual decrease."
- "Capital expenditure has increased sharply, indicating infrastructure focus."

Feature 9: Prediction Layer

The system will use historical budget data to forecast future trends.

Example for defense sector:

- X% increase expected based on previous trend
- Y% of total budget allocated
- Possible future spending direction
- New policy or scheme impact

Example for agriculture sector:

- New schemes introduced
- X% increase from last year
- Y% share of total budget
- Expected future allocation trend

The prediction layer will not claim exact future government decisions. It will provide trend-based estimates using past allocation data.

Feature 10: Foreign Budget Comparison

The system will compare India's budget with foreign budgets, especially the US Federal Budget.

Comparison areas:

- Education expenditure
- Healthcare expenditure
- Defense expenditure
- Finance sector support
- Private sector and business support
- Debt handling
- Climate and environment spending
- AI and technology investment
- Social security models

This feature will make the project more advanced and globally relevant.

Feature 11: Budget Speech vs Actual Allocation Comparison

The system will analyze budget speeches and compare stated policy priorities with actual fund allocation.

Example:

If the budget speech focuses heavily on agriculture, the system will check whether agriculture allocation has actually increased.

This feature helps identify whether government policy focus matches spending decisions.

7. Data Sources

The project will use official and reliable data sources such as:

- Ministry of Finance Budget Documents
- Union Budget PDFs
- Annual Financial Statement
- Budget at a Glance
- Expenditure Budget
- Receipts Budget
- Finance Bill
- Economic Survey
- RBI Handbook of Statistics
- Open Government Data Platform
- US Federal Budget data for foreign comparison

8. Key Variables to Include

The dataset should include:

- Total revenue
- Total expenditure
- Capital expenditure
- Revenue expenditure
- Fiscal deficit as percentage of GDP
- Primary deficit
- Tax revenue
- Non-tax revenue
- Borrowings
- Corporate tax
- Income tax
- GST collection
- Customs duty
- Sector-wise allocation
- GDP growth assumptions
- Inflation assumptions
- Actual spending vs budget estimate
- Revised estimate vs budget estimate

9. Machine Learning Usage

Machine learning should be used carefully and practically. The project should not overclaim that it can predict government decisions perfectly.

ML can be used for:

- Trend detection
- Sector-wise growth prediction
- Pattern clustering
- Anomaly detection
- Change-point detection
- Topic modeling on budget speeches
- Similarity comparison between Indian and foreign policies
- Forecasting future allocation trends

10. Minimal Working Project

The first version of the project should be simple but functional.

MVP Features

The minimal working project will include:

1. Upload Union Budget PDF.
2. Extract text and tables from the PDF.
3. Summarize the budget document.
4. Show government revenue sources using pie charts.
5. Show sector-wise allocation using bar graphs.
6. Compare current year allocation with previous year data.
7. Provide basic natural language question answering.
8. Generate a simple report with key insights.

MVP Output

The MVP should generate:

- Budget summary
- Sector-wise allocation table
- Revenue source chart
- Expenditure chart
- Previous year comparison
- Basic insights
- Simple Q&A responses

11. Sequential Feature Addition Plan

Phase 1: Data Collection and Cleaning

Collect budget data from official sources and prepare structured datasets.

Tasks:

- Download Union Budget PDFs.
- Collect sector-wise allocation data.
- Collect revenue and expenditure data.
- Clean and organize data year-wise.
- Store data in CSV or database format.

Phase 2: Basic Dashboard

Build a simple dashboard to show budget data.

Tasks:

- Create sector-wise allocation charts.
- Create government revenue pie chart.

- Create expenditure comparison graphs.
- Add year selection option.
- Add separate sections for education, health, defense, agriculture, finance, and private sector.

Phase 3: PDF Summarizer

Add document summarization.

Tasks:

- Upload budget PDF.
- Extract text from PDF.
- Generate summary.
- Highlight key tax reforms, schemes, and policy changes.

Phase 4: Previous Year Comparison

Add comparison with historical data.

Tasks:

- Compare current year with previous year.
- Calculate percentage increase or decrease.
- Compare sector-wise allocation.
- Compare budget estimate, revised estimate, and actual spending.

Phase 5: Insight Generation

Add rule-based and AI-based insights.

Tasks:

- Identify highest increase and decrease.
- Identify priority sectors.
- Explain who benefits.
- Explain who may be affected.
- Generate public-friendly explanations.

Phase 6: Machine Learning Layer

Add ML-based analysis.

Tasks:

- Use anomaly detection for unusual allocation changes.
- Use clustering to group similar sector trends.
- Use time-series forecasting for future allocation trends.
- Use topic modeling on budget speeches.

Phase 7: Foreign Budget Comparison

Add comparison with US budget data.

Tasks:

- Collect US Federal Budget sector-wise data.
- Compare education, health, defense, finance, and private sector spending.
- Show India vs US sector-wise expenditure charts.
- Explain policy differences.

Phase 8: Advanced Policy Intelligence

Add deeper policy analysis.

Tasks:

- Compare budget speech priorities with actual allocation.
- Track scheme-level allocation changes.
- Analyze inflation-adjusted spending.
- Compare capex and revenue expenditure impact.
- Generate final AI-based budget intelligence report.

12. Suggested Tech Stack

Frontend

- Streamlit

Backend

- Python
- FastAPI, optional for API-based architecture

Data Processing

- Pandas
- NumPy
- Tabula or Camelot for table extraction
- PyPDF2 or pdfplumber for PDF extraction

Machine Learning

- Scikit-learn
- Prophet or ARIMA for forecasting
- Sentence Transformers for semantic search
- LDA or BERTopic for topic modeling

AI and NLP

- LangChain
- Large Language Model API or local LLM
- RAG-based document question answering

Visualization

- Matplotlib
- Plotly
- Seaborn

Database

- MySQL or PostgreSQL
- Vector database for semantic search, if RAG is added

13. Expected Final Output

The final system will act as an AI-powered budget assistant where users can upload budget documents, ask questions, view visualizations, compare previous budgets, understand policy impact, and generate intelligent budget reports.

Example final outputs:

- "Education allocation increased by X% compared to last year."
- "Healthcare spending is below the selected benchmark."
- "Defense received Y% of the total budget."
- "Capital expenditure has increased, showing infrastructure focus."
- "The budget speech mentions agriculture strongly, but actual allocation increased only slightly."
- "Compared to the US, India spends a lower percentage on healthcare but a higher relative share on selected infrastructure areas."

14. Innovation of the Project

This project is innovative because it is not just a budget dashboard. Government websites already provide budget documents, graphs, and basic comparisons. However, this project adds intelligence on top of the data.

The innovation comes from:

- Natural language question answering
- Public-friendly budget explanation
- Sector-wise impact analysis
- Scheme-level comparison
- Anomaly detection
- Prediction of budget trends

- India vs foreign budget comparison
- Budget speech vs actual allocation analysis
- “Who benefits” and “who may be affected” explanation

Because of these features, the project becomes an AI-powered policy and budget intelligence system rather than a simple visualization tool.

15. Final Project Summary

Policy Bill Budget AI is a machine learning and AI-based Union Budget analysis system for India. It will help users understand how government money is collected, where taxpayer money is spent, and how budget priorities change over time. The system will analyze Union Budget PDFs, Finance Bills, policy documents, and sector-wise allocation data. It will compare current and previous budgets, generate visualizations, detect trends, identify unusual changes, forecast future allocation patterns, and compare Indian budget priorities with foreign budgets such as the US Federal Budget.

The project will be developed step by step, starting with a minimal working dashboard and PDF summarizer. Later, advanced features such as machine learning-based trend detection, anomaly detection, foreign policy comparison, and budget speech vs allocation analysis will be added. This makes the project practical, scalable, and more innovative than a normal budget comparison dashboard.